
Direct Preference Optimization: Your Language Model is Secretly a Reward Model

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Motivation

Limitation of Existing RLHF

1. Complex Multi-stage Pipeline

- 기존 RLHF는 여러 단계를 거치는 복잡한 파이프라인
 - Stage 1. SFT → Stage 2. Reward Modeling → Stage 3. RL optimization
- Reward Model과 Policy를 별도로 학습해야 하므로, 여러 모델과 학습 stage를 관리해야 하는 부담

2. Online Sampling during RL Fine-tuning

- PPO RL 단계에서 아래의 학습루프 반복
 - Repeated response sampling ($y \sim \pi_{\theta}(y|x)$) → Reward Model → Policy update
- LM generation 반복되어 cost ↑

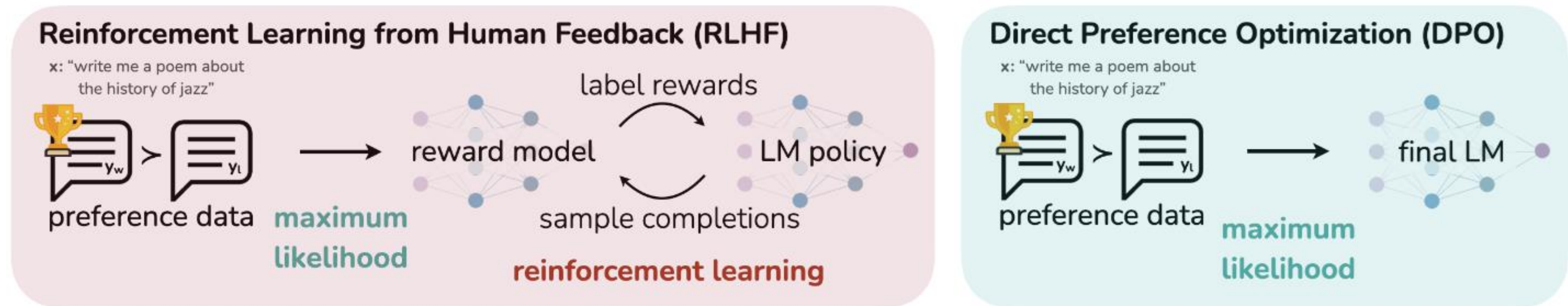
3. Unstable & Sensitive RL Optimization

- LLM RL optimization은 기본적으로 학습이 복잡하고 불안정
- KL coefficient 등 여러 sensitive hyperparameter tuning

Motivation

Can we learn directly from preferences without an explicit RM or RL?

DPO: Direct Preference Optimization



핵심: reward와 optimal policy 사이의 분석적 대응 관계

Reward를 policy로 재표현하고, Bradley–Terry 선호 모델에 대입
별도 RM 학습과 RL rollout 없이 preference pair로 policy를 직접 학습

Method

Preliminaries - RLHF objective

Reinforcement Learning from Human Feedback (RLHF) :

$$\max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(y|x)} [r_{\phi}(x, y)] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(y | x) || \pi_{\text{ref}}(y | x)]$$

참고: $\pi_{\text{ref}} = \pi_{\text{SFT}}$ (fixed)

1. Reward를 높이고 싶다
2. 하지만 원래 모델에서 너무 멀어지면 안된다

즉, RLHF의 목표는 “**High Reward + Stay close to SFT model**”

Bradley-Terry

어떤 데이터 (x, y_w, y_l) 는

- x : prompt
- y_w : preferred response (winner)
- y_l : rejected response (loser)

사람이 y_w 를 y_l 보다 선호할 확률은

- $P(y_w > y_l | x) = \sigma(r(x, y_w) - r(x, y_l))$: **Bradley-Terry preference model**
- 여기서 중요한 것은 절대 reward가 아니라 **reward의 차이**

결국 Reward Model 학습은 $r(x, y_w) > r(x, y_l)$ 가 되도록 만드는 binary classification

- $L_R = -\log \sigma(r(x, y_w) - r(x, y_l))$

Optimal Policy & Reward Reparameterization

Previous : $(x, y_w, y_l) \rightarrow r(x, y)$ via RM $\rightarrow \pi_\theta$ via PPO

Question : “Reward Model(RM)을 굳이 따로 만들고 PPO로 다시 최적화해야 하는가?”

$J_x(\pi) = -\beta D_{KL}(\pi \parallel \pi_r) + \beta \log Z(x)$ 따라서 최대값은 $\pi = \pi_r$ 에서 달성

앞선 RLHF objective의 optimal solution (최적해) :

$$\pi_r(y \mid x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp \left(\frac{1}{\beta} r(x, y) \right) \quad Z(x) = \sum_y \pi_{\text{ref}}(y \mid x) \exp \left(\frac{1}{\beta} r(x, y) \right)$$

- $Z(x)$ 는 모든 probability 합을 1로 만들어주는 normalization constant / partition function

위의 최적해에 $\log$ 를 취하고 식을 전개하면

$$r(x, y) = \beta \log \frac{\pi_r(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x)$$

Implicit Reward

$$r(x, y) = \beta \log \frac{\pi_r(y | x)}{\pi_{\text{ref}}(y | x)} + \beta \log Z(x).$$

Implicit reward: $\hat{r}_\theta(\mathbf{x}, y) = \beta \log [\pi_\theta(y|\mathbf{x}) / \pi_{\text{ref}}(y|\mathbf{x})]$

Reference 대비 response의 확률을 얼마나 높이거나 낮추었는가?

$\pi_\theta(y|\mathbf{x}) > \pi_{\text{ref}}(y|\mathbf{x})$ 해당 response의 확률을 reference 대비 높임

$\pi_\theta(y|\mathbf{x}) < \pi_{\text{ref}}(y|\mathbf{x})$ 해당 response의 확률을 reference 대비 낮춤

“Your Language Model is Secretly a Reward Model”

Policy likelihood ratio가 reward를 표현하므로, 별도 explicit RM이 필요 없음

Reward Equivalence

$$r(x, y) = \beta \log \frac{\pi_r(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x).$$

Reward equivalence: $r(x, y) - r'(x, y) = f(x)$

동일한 prompt의 모든 response에 같은 상수 $f(x)$ 를 더하거나 빼면

1. Bradley–Terry: reward 차이가 같으므로 preference distribution도 동일
2. KL-regularized RLHF: optimal policy도 동일

$\beta \log Z(x)$ 는 y 와 무관하므로 제거해도 같은 reward equivalence class

따라서 $\hat{r}_\theta(x, y) = \beta \log [\pi_\theta(y|x) / \pi_{\text{ref}}(y|x)]$ 만으로 충분

Reward in Bradley-Terry

$$r(x, y) = \beta \log \frac{\pi_r(y | x)}{\pi_{\text{ref}}(y | x)} + \beta \log Z(x)$$

기존 preference model $\mathbf{P}(\mathbf{y}_w > \mathbf{y}_l | \mathbf{x}) = \sigma(\mathbf{r}(\mathbf{x}, \mathbf{y}_w) - \mathbf{r}(\mathbf{x}, \mathbf{y}_l))$ 에 앞서 구한 Reward 수식을 대입

1. Winner Reward $\mathbf{r}(\mathbf{x}, \mathbf{y}_w) : \beta \log(\pi_{\theta}(\mathbf{y}_w | \mathbf{x}) / \pi_{\text{ref}}(\mathbf{y}_w | \mathbf{x})) + \beta \log Z(\mathbf{x})$
2. Loser Reward $\mathbf{r}(\mathbf{x}, \mathbf{y}_l) : \beta \log(\pi_{\theta}(\mathbf{y}_l | \mathbf{x}) / \pi_{\text{ref}}(\mathbf{y}_l | \mathbf{x})) + \beta \log Z(\mathbf{x})$

이 둘을 빼면,

- $\mathbf{r}(\mathbf{x}, \mathbf{y}_w) - \mathbf{r}(\mathbf{x}, \mathbf{y}_l) = \beta \log(\pi_{\theta}(\mathbf{y}_w | \mathbf{x}) / \pi_{\text{ref}}(\mathbf{y}_w | \mathbf{x})) - \beta \log(\pi_{\theta}(\mathbf{y}_l | \mathbf{x}) / \pi_{\text{ref}}(\mathbf{y}_l | \mathbf{x}))$
- 두 reward의 difference만 사용하기 때문에 $Z(\mathbf{x})$ 가 소거됨

최종적으로 $r(x, y_w) - r(x, y_l)$ 를 기존 preference model에 대입하면,

Final DPO Objective

Human Preference Probability를 명시적인 Reward 없이 Policy만으로 표현 가능

$$p^*(y_1 \succ y_2 \mid x) = \frac{1}{1 + \exp \left(\beta \log \frac{\pi^*(y_2 \mid x)}{\pi_{\text{ref}}(y_2 \mid x)} - \beta \log \frac{\pi^*(y_1 \mid x)}{\pi_{\text{ref}}(y_1 \mid x)} \right)}$$

정답 label은 항상 $y_w > y_l$ 이므로, 위 수식의 negative log likelihood를 최소화

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \right) \right]$$

Increase the relative preference for the **chosen response** over the **rejected response**, compared with the reference policy

- 단순히 $\pi(y_w) \uparrow$, $\pi(y_l) \downarrow$ 만 하는 것이 아니라,
- $\log(\pi_\theta(y_w \mid x) / \pi_{\text{ref}}(y_w \mid x)) > \log(\pi_\theta(y_l \mid x) / \pi_{\text{ref}}(y_l \mid x))$ 를 비교

β : RLHF의 KL penalty 계수 (fixed reward에서 β 가 클수록 reference에 가까운 최적해)

DPO Objective Gradient

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$$
$$\nabla_{\theta} \mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) =$$
$$- \beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\underbrace{\sigma(\hat{r}_{\theta}(x, y_l) - \hat{r}_{\theta}(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \left[\underbrace{\nabla_{\theta} \log \pi(y_w | x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_{\theta} \log \pi(y_l | x)}_{\text{decrease likelihood of } y_l} \right] \right]$$

$$\hat{r}_{\theta}(x, y) = \beta \log [\pi_{\theta}(y|x) / \pi_{\text{ref}}(y|x)]$$

Case 1 $\hat{r}_{\theta}(x, y_w) \gg \hat{r}_{\theta}(x, y_l)$

$\sigma(\hat{r}_l - \hat{r}_w) \approx 0$: 이미 올바르게 구분한 pair에는 작은 gradient

Case 2 $\hat{r}_{\theta}(x, y_w) \ll \hat{r}_{\theta}(x, y_l)$

$\sigma(\hat{r}_l - \hat{r}_w) \approx 1$: 잘못 구분한 pair에는 큰 gradient

Pair별 dynamic weighting이 핵심이며, 고정 weight를 쓰면 degeneration 가능

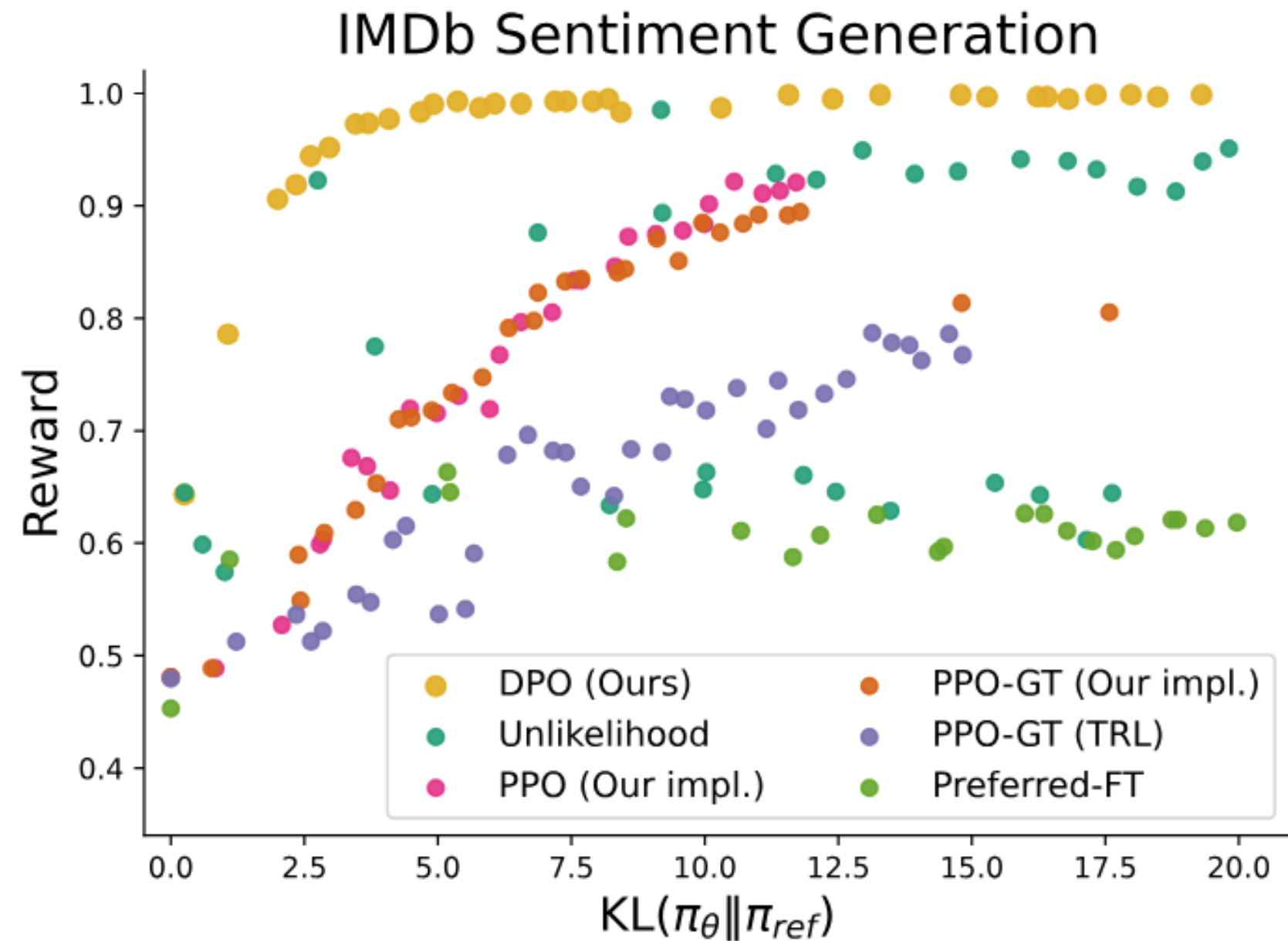
Experiment Setup

Task / Model	Goal	Evaluation
IMDb / GPT-2-large	Positive sentiment generation	Classifier reward vs. KL
Reddit TL;DR / GPT-J 6B	Summarization	GPT-4 / human evaluation
Anthropic HH / Pythia 2.8B	Single-turn dialogue	GPT-4 vs. chosen response

Baselines: SFT / Preferred-FT, Unlikelihood, PPO, Best of N

PPO-GT: IMDb에서 ground-truth sentiment classifier reward로 학습한 PPO

Reward-KL Trade-off



IMDb controlled sentiment

좌상단이 좋음

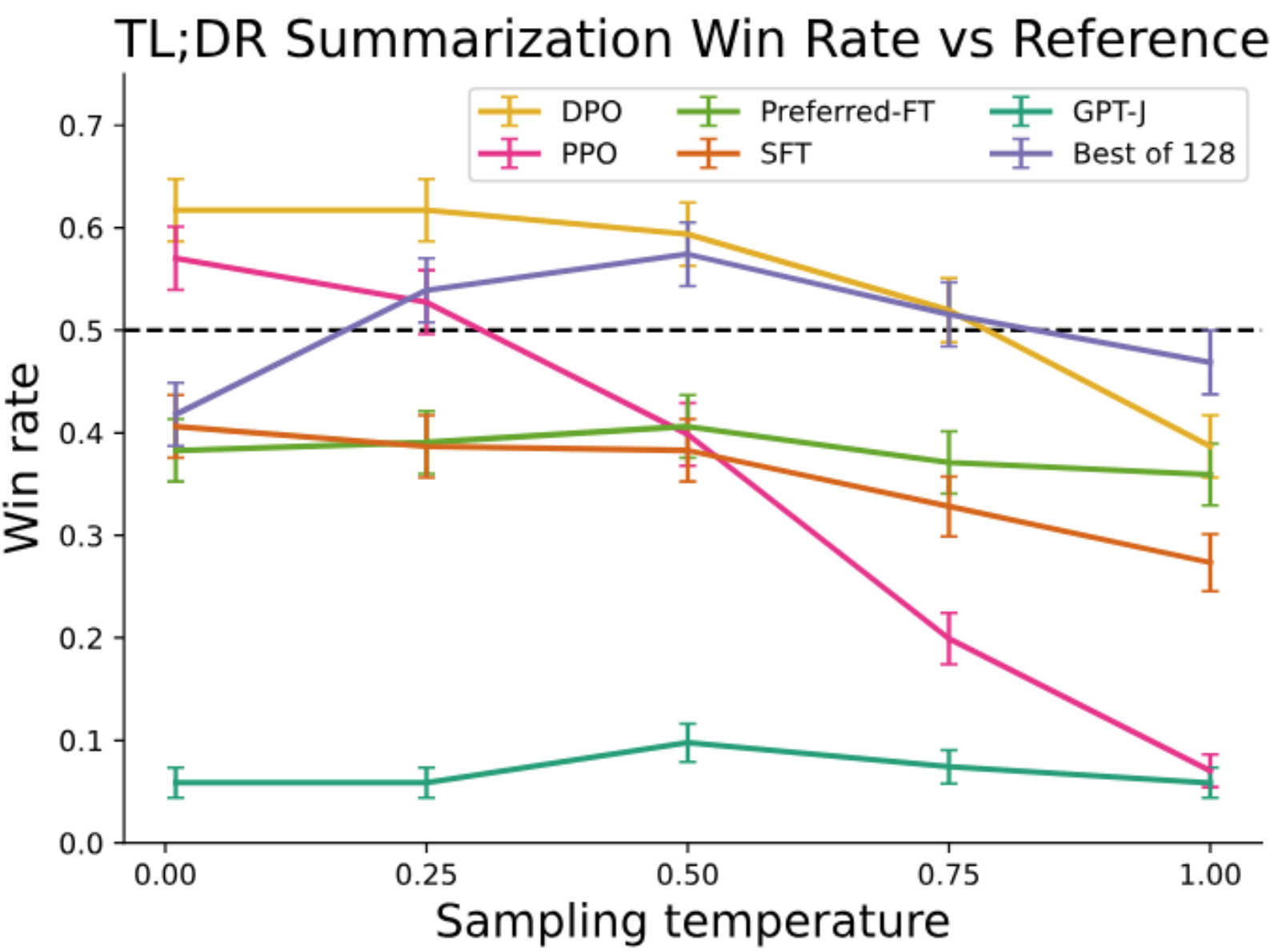
높은 true reward + 작은 KL deviation

DPO의 Reward-KL frontier가 PPO와 PPO-GT보다 우수

PPO-GT: ground-truth sentiment
classifier reward로 직접 학습한 PPO

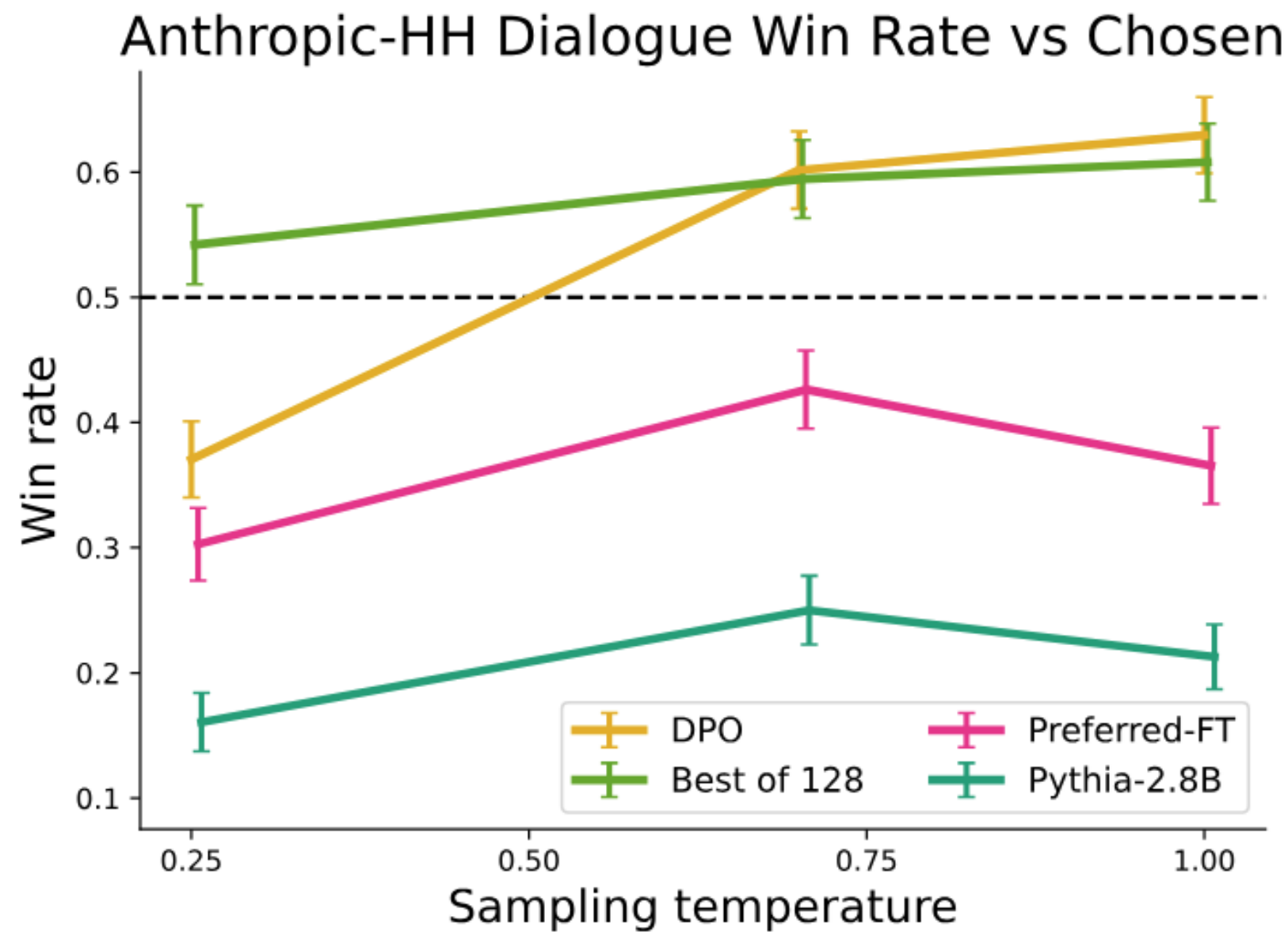
이 결과는 IMDb의 controlled experiment에 해당

Real Preference Tasks



TL;DR: GPT-4 win rate vs. human reference (T = 0)

DPO $\approx$ 61% vs. PPO $\approx$ 57%



HH: GPT-4 win rate vs. chosen response

Human eval (TL;DR): DPO (T = 0.25) preferred over PPO (T = 0) in 58% of judgments

Limitations & Discussion

Offline preference optimization

고정된 preference pair로 학습하며, 학습 중 새로운 response 탐색은 없음
Preference data의 coverage와 Bradley–Terry 모델 가정에 의존

Original paper: limitations & open questions

OOD 실험은 초기 근거이며, 더 폭넓은 일반화 검증이 필요
Unlabeled prompt 활용과 DPO의 reward over-optimization은 미해결 질문
실험 모델은 최대 6B이며, GPT-4 평가 결과는 judge prompt에 민감

Trade-off: 간단한 학습 파이프라인과 offline data의 한계

Conclusion

DPO learns directly from preferences with an implicit reward model

Reward와 optimal policy 사이의 분석적 대응을 이용하여
KL-regularized RLHF를 policy likelihood ratio의 classification loss로 재표현

$$\text{RLHF: } \max_{\pi} E_x [E_{y \sim \pi} [r(x, y)] - \beta D_{\text{KL}}(\pi \parallel \pi_{\text{ref}})]$$

$$\text{DPO: } L = -E_D [\log \sigma(\hat{r}_{\theta}(x, y_w) - \hat{r}_{\theta}(x, y_l))]$$

$$\hat{r}_{\theta}(x, y) = \beta \log [\pi_{\theta}(y|x) / \pi_{\text{ref}}(y|x)]$$

**Competitive or better results on the paper's benchmarks,
with a simpler training pipeline**